MILITARY SPECIFICATION

CARTRIDGE, 81MM, ILLUMINATING XM512E1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Illuminating Cartridge designated as XM512E1 for use with the 81MM Mortar.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-P-46568 - Primer, Percussion, M71E1, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1315

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DRAWINGS

ORDNANCE CORPS

8882162 - Cartridge, 81MM, Illuminating, XM512E1
8885050 - Container, Ammunition, Plastic, M- ,
for Cartridge, 81MM, Illuminating, XM512E1.

PUBLICATIONS

ORDNANCE CORPS

IEL-8882162 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Loaded cartridge.-The loaded cartridge shall comply with all requirements specified on Drawing (Dwg.) 8882162, and with all requirements specified in applicable specifications.

3.3 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.3.1 Failure to fire.-The cartridge shall not fail to fire.

3.3.2 Fuze.-The fuze shall function and cause ejection of the projectile illuminating assembly and the tail cone.

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3.3.3 Operation of the parachute.-The parachute shall open after ejection of the illuminating assembly and suspend the illuminant assembly during its period of burning.

3.3.4 Burning of illuminant.-The illuminant shall ignite and burn for not less than 80 seconds.

3.3.5 Premature burst.-There shall be no premature burst or metal parts failure in the barrel of the mortar or within 200 yards horizontal range of the mortar.

3.3.6 Horizontal range.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.4 Waterproofness (applicable to plastic container).-The plastic container shall be waterproof as evidenced by absence of air leakage when tested as specified in 4.3.1.

3.5 Workmanship

3.5.1 All processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.5.2 The painting and marking shall be restored by the loading and assembling contractor on any components furnished to him painted and marked when such painting and marking has been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the

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examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analyses on all material purchased directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the

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Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of loaded cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable). Each lot shall contain:

a. Loaded projectiles (including fins and increment holders) from one lot interfix number from one manufacturer.

b. Loaded ignition cartridges from one lot interfix number from one manufacturer.

c. Loaded primers from one lot interfix number from one manufacturer.

d. Loaded fuzes from one lot interfix number from one manufacturer.

e. Propellant increments made from not more than one lot of propellant from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105D except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Cartridge assembly, prior to assembly of fuze, primer, and fin assembly (see dwg. 8882162).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Ignition cartridge missing or inverted	Visual	01001
Major:	None defined		
Minor:	AQL 1.00 percent		
201.	Threads damaged	Visual	01002
202.	Evidence of poor workmanship (see 3.5)	Visual	01003

4.2.2.2 Cartridge assembly (see dwg. 8882162).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer loose or above flush with rear face of fin relief	Manual-Straight-edge	02001
2.	Fuze safety wire missing or insecure	Visual	02002
3.	Fin broken or distorted	Visual	02003
4.	Propellant increment missing	Visual	02004
Major:	AQL 0.65 percent		
101.	Torque test failure of fin	Test	02005
102.	Torque test failure of tail cone	Test	02006
103.	Torque test failure of fuze	Test	02007
104.	Total length, maximum (max)	Gage	02008
105.	Any component damaged or incorrectly assembled (except as described by Critical defect 3 above	Visual	02009

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Minor:	AQL 1.50 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	02010
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	02011
203.	Fuze not fully seated	Visual	02012
204.	Evidence of poor workmanship (see 3.5)	Visual	02013

4.2.2.3 Unsealed plastic container (see dwg 8885050).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Propellant increment cut or torn	Visual	03001
102.	O-Ring damaged or missing	Visual	03002
103.	Packing stop missing	Visual	03003
104.	Filler-disc assembly missing	Visual	03004
105.	Packing component missing	Visual	03005
Minor:	None defined		

4.2.2.4 Sealed plastic container, when applicable (see dwg. 8885050).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Container damaged	Visual	04001
102.	Seal incomplete or badly wrinkled	Visual	04002
103.	Closure loose	Manual	04003

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Minor:	AQL 1.50 percent		
201.	Seal tab missing.....	Visual	04004
202.	Marking misleading or unidentifiable.....	Visual	04005
203.	Contents loose.....	Manual	04006

4.2.2.5 Wrap around crate for plastic containers (see dwg. 8864663).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	DOD symbol missing or incorrect.....	Visual	05001
102.	Board broken or split.....	Visual	05002
103.	Containers incorrectly placed.....	Visual	05003
104.	Wire missing, broken or loose.....	Visual-Manual	05004
Minor:	AQL 1.50 percent		
201.	Marking misleading or unidentifiable.....	Visual	05005
202.	Wire loop improperly engaged.....	Visual-Manual	05006
203.	Car seal missing, unsealed or improperly positioned.....	Visual-Manual	05007
204.	Contents loose.....	Visual-Manual	05008

4.2.2.6 Unit load, strapped, applicable to wrap around crate (see dwg. ORDJU-4021).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:		AQL 0.25 percent	
101.	Pallet damaged	Visual	06001
102.	Strapping missing, loose or obviously mislocated	Visual	06002
Minor:		AQL 1.00 percent	
201.	Number of containers incorrect	Visual	06003
202.	Strapping improperly engaged	Visual	06004

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Primer, Percussion, M71E1 (see Specification MIL-P-46568), Code No. 02021. If the total elapsed time between original acceptance of the M71E1 primer lot and the assembly of that lot into complete round cartridges exceeds two years,or if the M71E1 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46568 immediately before the M71E1 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridge (see 6.4).

4.2.3.2 Profile and alignment (see dwg. 8882162) Major defect - Code No. 02022.-The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the requirement specified shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.3 Heat seal at joint (see dwg. 8865610) Major defect Code No. 02023.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105D, Table III-A, Code letter H and AQL 1.0 percent. The test shall be performed as specified on dwg. 8865610, using test equipment in accordance with 4.2.4.

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4.2.3.4 Waterproofness of plastic container (see 3.4) Major defect, Code No. 02024.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105D, Table III-A, code letter H and AQL 1.0 percent. The test shall be performed as specified in 4.3.1, using test equipment in accordance with 4.2.4.

4.2.3.5 Proving ground.-The cartridges shall be observed for compliance with the requirements as classified in Table I.

TABLE I

Defect	Defect Classification	Code No.
Premature burst or parts failure at less than 200 yards horizontal range (see 3.3.5)	Critical	02014
Cartridge fails to fire (see 3.3.1)	Major	02015
Fuze fails to function and eject illuminating assembly (see 3.3.2)	Major	02016
Parachute fails to open and suspend illuminant at proper rate of descent (see 3.3.3)	Major	02017
Illuminant fails to ignite or burns less than 60 seconds (see 3.3.4)	Major	02018
Range less than 80 percent of average horizontal range (see 3.3.6)	Major	02019
Illuminant burns at least 60 seconds but less than 80 seconds (see 3.3.4)	Minor	02020

4.2.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, the sampling plan for this test shall be in accordance with Standard MIL-STD-105D, Table II-A, Code letter J, utilizing an AQL of 2.50 percent for each Major defect and an AQL of 4.00 percent for each Minor defect. The lot shall be rejected if a critical defect occurs during any portion of the test, or the number of defectives equals or exceeds the rejection number specified. The test shall be performed as specified in 4.3.2.

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4.2.3.5.2 After three consecutive lots have met the criteria of 4.2.3.4.1, the sampling plan shall be in accordance with Standard MIL-STD-105D, Table II-A, Code letter G, utilizing an AQL of 2.50 percent for each Major defect and an AQL of 4.00 percent for each Minor defect. The lot shall be rejected if a critical defect occurs or the number of defectives equals or exceeds the rejection number specified. The test shall be performed as specified in 4.3.2.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 8882162 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Waterproofness of plastic container.-The sample container shall be tested for leakage by submerging in water contained in a vacuum vessel and drawing a vacuum. A vacuum differential of 216MM (8 1/2 inches) of mercury shall be held for 30 seconds (minimum). Observation for evaluation of air bubbles shall be made. A steady stream of recurring succession

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of bubbles from any surface or area of the closure seal shall be cause for rejection.

4.3.2 Proving ground.-The test shall be performed at a Government proving ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. The cartridge shall be fired from an 81MM Mortar, for which the round is standard. Eight "B" and one "A" propellant increments shall be used for each cartridge for this test. Observation shall be made for compliance with the requirements specified and, in addition, the burning time shall be recorded.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 8885050.

5.2 Marking

5.2.1 Marking of the plastic container shall be in accordance with dwg. 8865610.

5.3 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control Drawing.

6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial

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equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be

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submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1315-A355

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